Week06 Version control Current File

Project

- jina-embeddings-v2-base-zh
- 01_SQL基础.py
- 02_ORM框架.py
- 03_Faiss基础.py
- 04_ES测试.py
- 05_ES基础.py
- 06_ES进阶.py
- 07_ES向量检索.py
- ES配置说明.txt
- labels.json
- library.db
- questions.json
- RAG101_01_读取数据集.py
- RAG101_02_RAG检索TFIDF.py
- RAG101_03_RAG检索BM25.py

RAG101_01_读取数据集.py | test.py | RAG101_03_RAG检索BM25.py | RAG101_04_RAG检索BERT.py | RAG101_

```python
questions = json.load(open("questions.json",encoding="utf-8"))
print("questions.json")  # 打印提示信息，显示加载的JSON文件名
print(questions[0])  # 打印第一个问题的内容

print("\n")  # 打印空行

# 打开PDF文件，单个pdf简单的内容解析，没有做格式识别
pdf = pdfplumber.open("汽车知识手册.pdf")# 以这个文件作为模拟的知识库
print("pages: ", len(pdf.pages))  # 打印提示信息，显示PDF文件的页数
print(pdf.pages[23].extract_text())  # 提取并打印第23页的文本内容，不包含格式
```

Run: RAG101_01_读取数据集 | test1

```
■ 遮阳板上的卡片/票据夹。 01无线充电板/置物板
■ 手套箱。
02杯托
■ 车门存储空间。
■ 眼镜盒。 03储物箱
■ 前副仪表台。
■ 前排中央扶手储物箱。 注意！
■ 切勿强行将尺寸不合适的容器卡入杯托。
24

Process finished with exit code 0
```

Week06 > RAG101_01_读取数据集.py 8:60 LF UTF-8 4 spaces D:\ProgramData\miniconda3\envs\pytorch_310

Week06 Version control Current File

Project

- 07_ES向量检索.py
- ES配置说明.txt
- labels.json
- library.db
- questions.json
- RAG101_01_读取数据集.py
- RAG101_02_RAG检索TFIDF.py
- RAG101_03_RAG检索BM25.py
- RAG101_04_RAG检索BERT.py
- RAG101_05_RAG检索BERT_Segment.py
- RAG101_06_RAG重排序.py
- RAG101_07_RAG多路召回.py
- RAG101_08_RAG问答.py
- submit_bge_retrieval_top1.json
- submit_bge_retrieval_top10.json

T_Segment.py | RAG101_06_RAG重排序.py | test1.py | RAG101_07_RAG多路召回.py | RAG101_08_RAG问答.py

```python
def ask_gpt(content):
            return response.json()
        except:
            max_error += 1
            continue

bge = json.load(open('submit_bge_sgement_retrieval_top10.json',encoding="utf-8"))
bm25 = json.load(open('submit_bm25_retrieval_top10.json',encoding="utf-8"))

questions = json.load(open("questions.json",encoding="utf-8"))
pdf = pdfplumber.open("汽车知识手册.pdf")
pdf_content_dict = {}
for page_idx in range(len(pdf.pages)):
    pdf_content_dict['page_' + str(page_idx + 1)] = pdf.pages[page_idx].extract_text()

fusion_result = []
```

Run RAG101_08_RAG问答

```
在中央显示屏中点击 ，进入驾驶员侧座椅通风控制界面。
01设置副驾驶员侧座椅通风强度及开关控制。
驾驶员/副驾驶员侧座椅通风分三级调节，点击控制开关后在"关-低-
01设置驾驶员侧座椅通风强度及开关控制。
中-高"之间循环。
在中央显示屏中点击 ，进入副驾驶员侧座椅通风控制界面。
使用Lynk&CoApp打开/关闭前排座椅通风
打开/关闭前排座椅通风图标，登录Lynk&CoApp，按下
该图标可以打开/关闭前排座椅通风。
设置前排座椅通风时间
在中央显示屏中唤起空调控制界面，然后点击 -舒适。
116
【主题图说明】
```

Week06 > RAG101_08_RAG问答.py 71:61 LF UTF-8 4 spaces D:\ProgramData\miniconda3\envs\pytorch_310

Week06 Version control Current File

Project

- 07_ES向量检索.py
- ES配置说明.txt
- labels.json
- library.db
- questions.json
- RAG101_01_读取数据集.py
- RAG101_02_RAG检索TFIDF.py
- RAG101_03_RAG检索BM25.py
- RAG101_04_RAG检索BERT.py
- RAG101_05_RAG检索BERT_Segment.py
- RAG101_06_RAG重排序.py
- RAG101_07_RAG多路召回.py
- RAG101_08_RAG问答.py
- submit_bge_retrieval_top1.json
- submit_bge_retrieval_top10.json

RAG101_05_RAG检索BERT_Segment.py | RAG101_06_RAG重排序.py | test1.py | RAG101_07_RAG多路召回.py

```python
import json
bge = json.load(open('submit_bge_sgement_retrieval_top10.json',encoding="utf-8"))
bm25 = json.load(open('submit_bm25_retrieval_top10.json',encoding="utf-8"))

fusion_result = []
k = 60
for q1, q2 in zip(bge, bm25):
    fusion_score = {}
    for idx, q in enumerate(q1['reference']):
        if q not in fusion_score:
            fusion_score[q] = 1 / (idx + k)
        else:
            fusion_score[q] += 1 / (idx + k)

    for idx, q in enumerate(q2['reference']):
        if q not in fusion_score:
```

Run RAG101_07_RAG多路召回

```
D:\ProgramData\miniconda3\envs\pytorch_310\python.exe D:\网盘下载\AI\第6周：RAG工程化实现\Week06\RAG101_07_RAG多路召回.py

Process finished with exit code 0
```

Week06 > RAG101_07_RAG多路召回.py 4:1 LF UTF-8 4 spaces D:\ProgramData\miniconda3\envs\pytorch_310

Week06 Version control Current File

Project

- RAG101_02_RAG检索TFIDF.py
- RAG101_03_RAG检索BM25.py
- RAG101_04_RAG检索BERT.py
- RAG101_05_RAG检索BERT_Segment.py
- RAG101_06_RAG重排序.py
- RAG101_07_RAG多路召回.py
- RAG101_08_RAG问答.py
- submit_bge_retrieval_top1.json
- submit_bge_retrieval_top10.json
- submit_bge_sgement_retrieval_top1.json
- submit_bge_sgement_retrieval_top10.json
- submit_bm25_retrieval_top1.json
- submit_bm25_retrieval_top10.json
- submit_fusion_bge+bm25_rerank_retrieval.json
- submit_fusion_bge+bm25_retrieval.json

RAG检索BERT.py | RAG101_02_RAG检索TFIDF.py | RAG101_05_RAG检索BERT_Segment.py | RAG101_06_RAG重排序.py

```python
tokenizer = AutoTokenizer.from_pretrained(model_path)
rerank_model = AutoModelForSequenceClassification.from_pretrained(model_path)
rerank_model.eval()

# 进行召回合并
bge = json.load(open('submit_bge_sgement_retrieval_top10.json', encoding="utf-8"))
bm25 = json.load(open('submit_bm25_retrieval_top10.json', encoding="utf-8"))

fusion_result = []
k = 60
for q1, q2 in zip(bge, bm25):
    print(len(fusion_result), len(bge))

    fusion_score = {}
    for idx, q in enumerate(q1['reference']):
        if q not in fusion_score:
```

Run RAG101_06_RAG重排序

```
D:\ProgramData\miniconda3\envs\pytorch_310\python.exe D:\网盘下载\AI\第6周：RAG工程化实现\Week06\RAG101_06_RAG重排序.py
Loading weights: 100%|██████████| 201/201 [00:00<00:00, 5120.97it/s]
XLMRobertaForSequenceClassification LOAD REPORT from: D:\网盘下载\AI\第6周：RAG工程化实现\Week06\bge-reranker-base
Key                            | Status     |  |
-------------------------------+------------+--+-
roberta.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
0 301
1 301
2 301
3 301
```

Week06 > RAG101_06_RAG重排序.py 35:46 LF UTF-8 4 spaces D:\ProgramData\miniconda3\envs\pytorch_310

Week06 Version control Current File

Project

- RAG101_02_RAG检索TFIDF.py
- RAG101_03_RAG检索BM25.py
- RAG101_04_RAG检索BERT.py
- RAG101_05_RAG检索BERT_Segment.py
- RAG101_06_RAG重排序.py
- RAG101_07_RAG多路召回.py
- RAG101_08_RAG问答.py
- submit_bge_retrieval_top1.json
- submit_bge_retrieval_top10.json
- submit_bge_sgement_retrieval_top1.json
- submit_bge_sgement_retrieval_top10.json
- submit_bm25_retrieval_top1.json
- submit_bm25_retrieval_top10.json
- submit_fusion_bge+bm25_rerank_retrieval.json
- submit_fusion_bge+bm25_retrieval.json

RAG101_04_RAG检索BERT.py | RAG101_02_RAG检索TFIDF.py | RAG101_05_RAG检索BERT_Segment.py | RAG101_06_R

```python
for page_idx in range(len(pdf.pages)):
    text = pdf.pages[page_idx].extract_text()
    for chunk_text in split_text_fixed_size(text, chunk_size: 40):
        pdf_content.append({
            'page': 'page_' + str(page_idx + 1),
            'content': chunk_text
        })

model = SentenceTransformer( model_name_or_path: './bge-small-zh-v1.5', local_files_only=True)
question_sentences = [x['question'] for x in questions]
pdf_content_sentences = [x['content'] for x in pdf_content]

question_embeddings = model.encode(question_sentences, normalize_embeddings=True, show_progress_bar=True)
pdf_embeddings = model.encode(pdf_content_sentences, normalize_embeddings=True, show_progress_bar=True)
```

Run RAG101_05_RAG检索BERT_Segment

```
BertModel LOAD REPORT from: ./bge-small-zh-v1.5
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Batches: 100%|██████████| 10/10 [00:01<00:00,  6.73it/s]
Batches: 100%|██████████| 117/117 [00:19<00:00,  6.03it/s]

Process finished with exit code 0
```

Week06 > RAG101_05_RAG检索BERT_Segment.py 33:74 LF UTF-8 4 spaces D:\ProgramData\miniconda3\envs\pytorch_310

Week06 Version control Current File

Project

- 06_ES进阶.py
- 07_ES向量检索.py
- ES配置说明.txt
- labels.json
- library.db
- questions.json
- RAG101_01_读取数据集.py
- RAG101_02_RAG检索TFIDF.py
- RAG101_03_RAG检索BM25.py
- RAG101_04_RAG检索BERT.py
- RAG101_05_RAG检索BERT_Segment.py
- RAG101_06_RAG重排序.py
- RAG101_07_RAG多路召回.py
- RAG101_08_RAG问答.py
- submit_bge_retrieval_top1.json

RAG101_04_RAG检索BERT.py | RAG101_02_RAG检索TFIDF.py | RAG101_05_RAG检索BERT_Segment.py | RAG101_06_R

```python
for query_idx, feat in enumerate(question_embeddings):
    score = feat @ pdf_embeddings.T
    max_score_page_idx = score.argsort()[::-1] + 1
    questions[query_idx]['reference'] = ['page_' + str(x) for x in max_score_page_idx[:10]]

with open('submit_bge_retrieval_top10.json', 'w', encoding='utf8') as up:
    json.dump(questions, up, ensure_ascii=False, indent=4)


# jinaai
# modelscope download --model jinaai/jina-embeddings-v2-base-zh --local_dir jinaai/jina-embeddings-v2-base-zh
model = SentenceTransformer( model_name_or_path: './bge-small-zh-v1.5', local_files_only=True)
question_sentences = [x['question'] for x in questions]
pdf_content_sentences = [x['content'] for x in pdf_content]
```

Run RAG101_04_RAG检索BERT

```
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Loading weights: 100%|██████████| 71/71 [00:00<00:00, 5408.56it/s]
BertModel LOAD REPORT from: ./bge-small-zh-v1.5
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Process finished with exit code 0
```

Week06 Version control Current File

Project

- 06_ES进阶.py
- 07_ES向量检索.py
- ES配置说明.txt
- labels.json
- library.db
- questions.json
- RAG101_01_读取数据集.py
- RAG101_02_RAG检索TFIDF.py
- RAG101_03_RAG检索BM25.py
- RAG101_04_RAG检索BERT.py
- RAG101_05_RAG检索BERT_Segment.py
- RAG101_06_RAG重排序.py
- RAG101_07_RAG多路召回.py
- RAG101_08_RAG问答.py
- submit_bge_retrieval_top1.json

集.py test.py RAG101_03_RAG检索BM25.py RAG101_04_RAG检索BERT.py RAG101_02_RAG检索TFIDF.py

```python
import json
import pdfplumber
import jieba
import numpy as np
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import normalize

#读取数据集
questions = json.load(open("questions.json", encoding="utf-8"))
pdf = pdfplumber.open("汽车知识手册.pdf")
pdf_content = []
for page_idx in range(len(pdf.pages)):
    pdf_content.append({
        'page': 'page_' + str(page_idx + 1),
        'content': pdf.pages[page_idx].extract_text()
    })
```

Run RAG101_03_RAG检索BM25 test1

```
D:\ProgramData\miniconda3\envs\pytorch_310\python.exe D:\网盘下载\AI\第6周：RAG工程化实现\Week06\RAG101_03_RAG检索BM25.py
D:\ProgramData\miniconda3\envs\pytorch_310\lib\site-packages\jieba\_compat.py:18: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/p
  import pkg_resources
Building prefix dict from the default dictionary ...
Loading model from cache C:\Users\86157\AppData\Local\Temp\jieba.cache
Loading model cost 0.620 seconds.
Prefix dict has been built successfully.

Process finished with exit code 0
```

Week06 > RAG101_03_RAG检索BM25.py 9:1 (63 chars) LF UTF-8 4 spaces D:\ProgramData\miniconda3\envs\pytorch_310

Week06 Version control Current File

Project

- 06_ES进阶.py
- 07_ES向量检索.py
- ES配置说明.txt
- labels.json
- library.db
- questions.json
- RAG101_01_读取数据集.py
- RAG101_02_RAG检索TFIDF.py
- RAG101_03_RAG检索BM25.py
- RAG101_04_RAG检索BERT.py
- RAG101_05_RAG检索BERT_Segment.py
- RAG101_06_RAG重排序.py
- RAG101_07_RAG多路召回.py
- RAG101_08_RAG问答.py
- submit_bge_retrieval_top1.json

集.py | test.py | RAG101_03_RAG检索BM25.py | RAG101_04_RAG检索BERT.py | RAG101_02_RAG检索TFIDF.py

```python
    score = feat @ pdf_content_feat.T
    score = score.toarray()[0]
    max_score_page_idx = score.argsort()[::-1] + 1

    # 用户提问 与 最相关的10个页面
    questions[query_idx]['reference'] = ['page_' + str(x) for x in max_score_page_idx[:10]]

with open('submit_tfidf_retrieval_top10.json', 'w', encoding='utf8') as up:
    json.dump(questions, up, ensure_ascii=False, indent=4)
```

Run: RAG101_02_RAG检索TFIDF | test1

```
D:\ProgramData\miniconda3\envs\pytorch_310\python.exe D:\网盘下载\AI\第6周：RAG工程化实现\Week06\RAG101_02_RAG检索TFIDF.py
D:\ProgramData\miniconda3\envs\pytorch_310\lib\site-packages\jieba\_compat.py:18: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/p
  import pkg_resources
Building prefix dict from the default dictionary ...
Loading model from cache C:\Users\86157\AppData\Local\Temp\jieba.cache
Loading model cost 0.599 seconds.
Prefix dict has been built successfully.

Process finished with exit code 0
```

Week06 > RAG101_02_RAG检索TFIDF.py 56:1 LF UTF-8 4 spaces D:\ProgramData\miniconda3\envs\pytorch_310